Joint focus issue by Nature Astronomy and Nature Physics on gravitational waves

The focus issue will consist of three parts, focusing on detectors, science and challenges for future observatories of gravitational waves.

Gravitational wave detectors

Retrospective on detectors (Review Article for Nature Astronomy)

This Review offers a retrospective on the development of gravitational wave detectors – including ground- and space- based interferometry and pulsar timing arrays – and examines the evolution of their functionalities and sensitivities.

Future gravitational wave detectors (Perspective Article for Nature Astronomy)

This Perspective offers an overview of future ground- and space-based interferometers and PTAs.

Gravitational wave science

Tests of general relativity and searches for new physics with gravitational waves (Review article for Nature Physics)

This review focuses on tests of general relativity and searches for new physics with gravitational wave observations.

Gravitational waves as probes of the early Universe (Perspective for Nature Physics)

Gravitational waves can tell us about the early Universe, such as inflation, phase transitions, and yield estimates of the Hubble constant and other cosmological parameters. This perspective article gives an overview of the current status of such studies, and an outlook of what we can learn in the future, including from pulsar timing arrays.

Population studies (Perspective article for Nature Physics)

With the growing number of gravitational wave events, population analyses can tell us more about, for example, the formation of compact objects and binary stellar evolution. This perspective article gives an overview of this emergent field.

Supermassive black holes (Review Article for Nature Astronomy)

Pulsar timing array measurements of the stochastic gravitational wave background contain a wealth of potential information about supermassive black hole (SMBH) physics, formation and evolution, and in the near future these will likely be augmented by detections of continuous gravitational waves from individual SMBH binaries. This Review provides an overview of the science and its complementarity with results from the Event Horizon Telescope.

Challenges for future gravitational wave observatories

Global fit (Perspective for Nature Astronomy)

Data from upcoming space-based detectors such as LISA and TianQin will contain thousands of overlapping sources, together with multiple complex sources of noise. This Perspective explores how such a technical challenge can feasibly be solved.

Multimessenger (Comment for Nature Astronomy)

With upcoming technological developments, the detection of mergers involving neutron stars may become commonplace, which increase demand for electromagnetic follow-up. Equally, localising black hole binary mergers using electromagnetic observations would substantially increase their scientific yield. This Comment outlines how such multimessenger follow-up can be efficiently coordinated at scale.

Waveform modelling for interferometers (Comment for Nature Physics)

For interferometric gravitational wave detectors, the modelling of the waveform is a major source of systematic uncertainty. This comment discusses how the modelling can be improved by applying approaches from other areas, for example, from quantum-field theory.

Noise and stochastic signal modelling for pulsar timing arrays (Comment for Nature Physics)

A major challenge for pulsar timing arrays is understanding stochastic signals and noise in the data. This requires incorporating approaches and results from communities beyond traditional pulsar timing array research, as this comment outlines.

Content Types at Nature Physics

A **review** is a balanced, high-level overview (around 4000–5000 words with up to 6 display items and a maximum of 100 references) of a particular field. A **perspective** (around 3000 words with up to 4 display items and no more than 50 references) either gives an overview of an emerging field or a technical, scholarly discussion that may advocate a particular point of view. **Comments** are intended as opinion pieces of 1000–2000 words with up to 10 references (display items are optional).